Hess's Law and Enthalpies of Formation

Combining equations and calculating ΔH from formation data

Name _____ Class _____ Date _____

STANDARD ENTHALPIES OF FORMATION, ΔHf° (kJ/mol, 25 °C)

$CO_2(g)$ −393.5 · $CO(g)$ −110.5 · $H_2O(l)$ −285.8 · $H_2O(g)$ −241.8 · $CH_4(g)$ −74.6

$C_2H_4(g)$ +52.4 · $C_2H_6(g)$ −84.0 · $C_3H_8(g)$ −103.8 · $C_2H_5OH(l)$ −277.6 · $C_6H_{12}O_6(s)$ −1273.3

$NH_3(g)$ −45.9 · $NO(g)$ +91.3 · $CaCO_3(s)$ −1207.6 · $CaO(s)$ −634.9 · $Fe_2O_3(s)$ −824.2 · $Al_2O_3(s)$ −1675.7

Elements in their standard states: 0 · $M(C_6H_{12}O_6)$ = 180.16 g/mol

ΔH° = Σ nΔHf°(products) − Σ nΔHf°(reactants). Show units in every step.

A. Understanding

1. State Hess's law and explain why it is true.

2. Which of these have ΔHf° = 0? $O_2(g)$, $O_3(g)$, $N_2(g)$, $H_2O(l)$, $Fe(s)$, $C(graphite)$, $CO_2(g)$

B. Combining equations

3. Given: $N_2(g) + O_2(g) \longrightarrow 2\,NO(g)$, ΔH = +182.6 kJ, and $2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$, ΔH = −116.2 kJ. (a) Find ΔH for $N_2(g) + 2\,O_2(g) \longrightarrow 2\,NO_2(g)$. (b) Hence find ΔHf° of $NO_2(g)$.

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C. Using enthalpies of formation

4. Calculate ΔH° for: (a) $C_2H_4\,(g) + H_2\,(g) \longrightarrow C_2H_6\,(g)$ (b) $CaCO_3\,(s) \longrightarrow CaO\,(s) + CO_2\,(g)$ (c) $Fe_2O_3\,(s) + 2\,Al\,(s) \longrightarrow Al_2O_3\,(s) + 2\,Fe\,(s)$ (the thermite reaction)

5. Calculate ΔH° for the combustion of propane: $C_3H_8\,(g) + 5\,O_2\,(g) \longrightarrow 3\,CO_2\,(g) + 4\,H_2O\,(l)$

6. Calculate ΔH° for $4\,NH_3\,(g) + 5\,O_2\,(g) \longrightarrow 4\,NO\,(g) + 6\,H_2O\,(g)$, the first step in making nitric acid.

7. Use the table to find ΔH° for $H_2O\,(l) \longrightarrow H_2O\,(g)$. What does this value represent?

D. Challenge

8. The combustion of ethanol, $C_2H_5OH\,(l) + 3\,O_2\,(g) \longrightarrow 2\,CO_2\,(g) + 3\,H_2O\,(l)$, has $\Delta H^\circ = -1366.8$ kJ. Use this and the table (ignoring the ethanol value) to calculate ΔH_f° of ethanol.

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9. Calculate ΔH° for the combustion of glucose, $C_6H_{12}O_6\,(s) + 6\,O_2\,(g) \longrightarrow 6\,CO_2\,(g) + 6\,H_2O\,(l)$, and the energy released when 10.0 g of glucose is oxidized in the body.

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